Segway

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1 Boundry Condition

$$\mathrm{E_K^{body}} = \frac{1}{2}\,\mathrm{m_b}\,(\dot{\mathrm{x}}_\mathrm{b}^2 + \dot{\mathrm{y}}_\mathrm{b}^2 + \dot{\mathrm{z}}_\mathrm{b}^2) + \frac{1}{2}\,\mathrm{I_{b_{yy}}}\,\dot{\theta}_\mathrm{b}^2 + \frac{1}{2}\,(\mathrm{I_{b_{xx}}}\sin^2(\theta_\mathrm{b}) + \mathrm{I_{b_{zz}}}\cos^2(\theta_\mathrm{b}))\,\dot{\varphi}_\mathrm{b}^2$$

$$\mathrm{E_K^{wheels}} = \frac{1}{2}\,\mathrm{m_w}\,(\dot{\mathrm{x}}_{\mathrm{w_{left}}}^2 + \dot{\mathrm{x}}_{\mathrm{w_{right}}}^2 + \dot{\mathrm{y}}_{\mathrm{w_{left}}}^2 + \dot{\mathrm{y}}_{\mathrm{w_{right}}}^2 + 2\,\dot{\mathrm{z}}_\mathrm{w}^2) + \frac{1}{2}\,\mathrm{I_w}\,(\dot{\theta}_{\mathrm{w_{left}}}^2 + \dot{\theta}_{\mathrm{w_{right}}}^2)$$

$$\mathrm{E_P^{body}} = \mathrm{m_b}\,\mathrm{g}\,(\mathrm{l_b}\cos(\theta_\mathrm{b}) + \mathrm{z_w})$$

$$\mathrm{E_P^{wheels}} = 2\,\mathrm{m_w}\,\mathrm{g}\,\mathrm{z_w}$$

$$\mathrm{x_b} = \mathrm{x_w} + \mathrm{x_0} + \mathrm{l_b}\sin(\theta_\mathrm{b})\cos(\varphi_\mathrm{b})$$

$$\mathrm{y_b} = \mathrm{y_w} + \mathrm{y_0} + \mathrm{l_b}\sin(\theta_\mathrm{b})\sin(\varphi_\mathrm{b})$$

$$\mathrm{z_b} = \mathrm{z_w} + \mathrm{z_0} + \mathrm{l_b}\cos(\theta_\mathrm{b})$$

$$\mathrm{x_{w_{left}}} = \mathrm{x_w} - \frac{\mathrm{L}}{2}\sin(\varphi_\mathrm{w})$$

$$\mathrm{y_{w_{left}}} = \mathrm{y_w} + \frac{\mathrm{L}}{2}\cos(\varphi_\mathrm{w})$$

$$\mathrm{x_{w_{right}}} = \mathrm{x_w} + \frac{\mathrm{L}}{2}\sin(\varphi_\mathrm{w})$$

$$\mathrm{y_{w_{right}}} = \mathrm{y_w} - \frac{\mathrm{L}}{2}\cos(\varphi_\mathrm{w})$$

$$\dot{\mathrm{x}}_{\mathrm{w}} = \frac{\mathrm{z}_{\mathrm{w}}}{2}\left(\dot{\theta}_{\mathrm{w}_{\mathrm{left}}} + \dot{\theta}_{\mathrm{w}_{\mathrm{right}}}\right)\cos(\varphi_{\mathrm{w}})$$

$$\dot{\mathrm{y}}_{\mathrm{w}} = \frac{\mathrm{z}_{\mathrm{w}}}{2}\left(\dot{\theta}_{\mathrm{w}_{\mathrm{left}}} + \dot{\theta}_{\mathrm{w}_{\mathrm{right}}}\right)\sin(\varphi_{\mathrm{w}})$$

$$\dot{\varphi}_{\mathrm{w}} = \frac{\mathrm{z}_{\mathrm{w}}}{\mathrm{L}}\left(\dot{\theta}_{\mathrm{w}_{\mathrm{right}}} - \dot{\theta}_{\mathrm{w}_{\mathrm{left}}}\right)$$

$$\mathrm{z}_{\mathrm{w}} \coloneqq \mathrm{l}_{\mathrm{w}}\ ,\ \mathrm{z}_0 \coloneqq 0\ ,\ \varphi \coloneqq \varphi_{\mathrm{b}} = \varphi_{\mathrm{w}}$$

$$\mathrm{E_K} \triangleq \mathrm{E_K^{body}} + \mathrm{E_K^{wheel}}\ ,\ \mathrm{E_P} \triangleq \mathrm{E_P^{body}} + \mathrm{E_P^{wheel}}\ ,\ \mathcal{L} \triangleq \mathrm{E_K} - \mathrm{E_P}$$

$$\begin{aligned} E_K = \frac{m_b}{2}\Big(&\frac{l_w^2}{4}\left(\dot{\theta}_{w_{left}} + \dot{\theta}_{w_{right}}\right)^2 + l_b^2\left(\dot{\theta}_b^2 + \right. \\ &\left. + \frac{l_w^2}{L^2}\left(\dot{\theta}_{w_{right}} - \dot{\theta}_{w_{left}}\right)^2 sin^2(\theta_b)\right)\Big) + \\ &+ \frac{m_b\, l_b\, l_w}{2}\left(\dot{\theta}_{w_{left}} + \dot{\theta}_{w_{right}}\right)\dot{\theta}_b\, cos(\theta_b) + \frac{I_b}{2}\,\dot{\theta}_b^2 + \\ &+ \frac{I_{b_{zz}}\, l_w^2}{2\, L^2}\left(\dot{\theta}_{w_{right}} - \dot{\theta}_{w_{left}}\right)^2 + \frac{m_w\, l_w^2}{2}\left(\dot{\theta}_{w_{left}}^2 + \dot{\theta}_{w_{right}}^2\right) + \\ &+ \frac{I_w}{2}\left(\dot{\theta}_{w_{left}}^2 + \dot{\theta}_{w_{right}}^2\right) \end{aligned}$$

$$\mathrm{E_P} = \mathrm{m_b\, g\left(l_b \cos(\theta_b) + l_w\right) + 2\, m_w\, g\, l_w}$$

$$\begin{aligned} &\partial_{t\,,\,\dot{\theta}_b}\mathcal{L} - \partial_{\theta_b}\mathcal{L} = Q_{\theta_b} \Rightarrow \\ &\quad m_b\, l_b^2\,\ddot{\theta}_b + \frac{m_b\, l_b\, l_w}{2}\left(\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}\right) cos(\theta_b) + I_b\,\ddot{\theta}_b + \\ &\quad - \frac{m_b\, l_b^2\, l_w^2}{L^2}\left(\dot{\theta}_{w_{right}} - \dot{\theta}_{w_{left}}\right)^2 sin(\theta_b)\, cos(\theta_b) + \\ &\quad - m_b\, g\, l_b\, sin(\theta_b) = Q_{\theta_b} \end{aligned}$$

$$\begin{aligned} &\partial_{t\,,\,\dot{\theta}_{w_{left}}}\mathcal{L} - \partial_{\theta_{w_{left}}}\mathcal{L} = Q_{\theta_{w_{left}}} \Rightarrow \\ &\quad m_b\, l_w^2\left(\frac{1}{4}\left(\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}\right) - \frac{l_b^2}{L^2}\left(\left(\ddot{\theta}_{w_{right}} - \ddot{\theta}_{w_{left}}\right) sin^2(\theta_b) + \right.\right. \\ &\quad \left.\left. 2\left(\dot{\theta}_{w_{right}} - \dot{\theta}_{w_{left}}\right)\dot{\theta}_b\, sin(\theta_b)\, cos(\theta_b)\right)\right) + \frac{m_b\, l_b\, l_w}{2}\left(\ddot{\theta}_b\, cos(\theta_b) + \right. \\ &\quad \left. - \dot{\theta_b}^{\,2}\, sin(\theta_b)\right) - \frac{I_{b_{zz}}\, l_w^2}{L^2}\left(\ddot{\theta}_{w_{right}} - \ddot{\theta}_{w_{left}}\right) + m_w\, l_w^2\,\ddot{\theta}_{w_{left}} + I_w\,\ddot{\theta}_{w_{left}} = Q_{\theta_{w_{left}}} \end{aligned}$$

$$\partial_{t\,,\,\dot{\theta}_{w_{right}}}\mathcal{L} - \partial_{\theta_{w_{right}}}\mathcal{L} = Q_{\theta_{w_{right}}} \Rightarrow$$

$$m_b\, l_w^2 \left(\frac{1}{4}\left(\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}\right) + \frac{l_b^2}{L^2}\left(\left(\ddot{\theta}_{w_{right}} - \ddot{\theta}_{w_{left}}\right) sin^2(\theta_b) + \right.\right.$$

$$\left.\left. 2\left(\dot{\theta}_{w_{right}} - \dot{\theta}_{w_{left}}\right)\dot{\theta}_b\, sin(\theta_b)\, cos(\theta_b)\right)\right) + \frac{m_b\, l_b\, l_w}{2}\left(\ddot{\theta}_b\, cos(\theta_b) + \right.$$

$$\left. - {\dot{\theta}_b}^2\, sin(\theta_b)\right) + \frac{I_{b_{zz}}\, l_w^2}{L^2}\left(\ddot{\theta}_{w_{right}} - \ddot{\theta}_{w_{left}}\right) + m_w\, l_w^2\, \ddot{\theta}_{w_{right}} + I_w\, \ddot{\theta}_{w_{right}} = Q_{\theta_{w_{right}}}$$